

Interconnecting Series and Parallel ODEs

Local equations, connection constraints and phasors

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2026-10-08

Abstract

Series and parallel ODE blocks connect through shared terminal variables and signed sums. We derive these constraints, assemble a mixed connection, and retain its four initial coordinates. Substitution of $D = i\omega$ gives the corresponding phasor equations. Exact elimination recovers the same result as a scalar ODE.

PDF created: 2026-10-08 11:27:16 UTC

Latest on GitHub: <https://github.com/hobnilre/physics-ode-interconnect-ser-par>

1 Local equations and connection rules

Use electrical connection names and let $D = d/dt$. A block has terminal voltage e_j and current i_j . Orient series voltage drops along the common current; orient parallel branch currents in the same direction between the common terminals. All L, R, C parameters below are positive constants.

Use the LRC and CRL forms derived in *The ODE Template and Its Domain Equivalents*, Section 2. For a series LRC block $\mathcal{S}$, charge q gives

$$e_s = L_s D^2 q + R_s Dq + \frac{q}{C_s}, \quad i_s = Dq. \quad (1)$$

For a parallel CRL block $\mathcal{P}$, flux linkage λ gives

$$i_p = C_p D^2 \lambda + \frac{D\lambda}{R_p} + \frac{\lambda}{L_p}, \quad e_p = D\lambda. \quad (2)$$

The terminal driving quantity of either ODE is supplied by the rest of the connected system.

For two blocks, a series connection imposes

$$\boxed{i_1 = i_2 = i, \quad e = e_1 + e_2,} \quad (3)$$

whereas a parallel connection imposes

$$\boxed{e_1 = e_2 = e, \quad i = i_1 + i_2.} \quad (4)$$

The same balances apply to any finite number of blocks.

Either block type can enter either connection. Apply the common-variable and sum constraints at each junction, retaining the internal coordinates.

2 A mixed interconnection

Connect $\mathcal{S}$ and $\mathcal{P}$ in series, with total voltage $e(t)$, common current $i(t)$ and internal voltage $e_p(t)$. Thus $e_s = e - e_p$ and $i_s = i_p = i$. Inserting these relations into (1)–(2) gives

$$L_s D^2 q + R_s Dq + \frac{q}{C_s} + D\lambda = e, \quad (5)$$

$$C_p D^2 \lambda + \frac{D\lambda}{R_p} + \frac{\lambda}{L_p} = Dq. \quad (6)$$

The term $D\lambda$ in the first equation is the common voltage of the parallel block. The term Dq in the second is the common current of the series block.

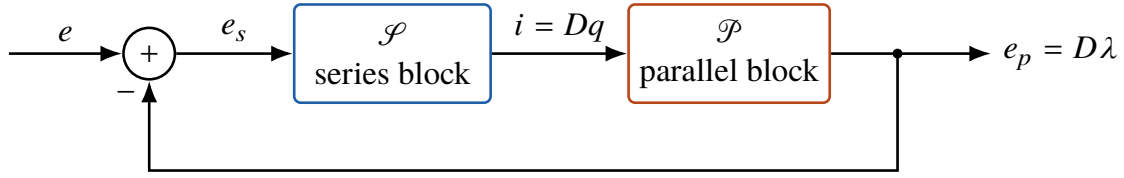


Figure 1. Schematic of the equation interconnection. The series block receives $e - e_p$ and determines i ; the parallel block receives i and determines e_p . The return path enforces $e_s = e - e_p$.

To retain all initial coordinates, put $i = Dq$ and $e_p = D\lambda$. The coupled equations become

$$\begin{aligned} Dq &= i, \\ L_s Di &= e - R_s i - \frac{q}{C_s} - e_p, \\ C_p De_p &= i - \frac{e_p}{R_p} - \frac{\lambda}{L_p}, \\ D\lambda &= e_p. \end{aligned} \quad (7)$$

This is an explicit system of four first-order ODEs for

$$x = (q, i, e_p, \lambda)^\top, \quad x(t_0) = (q_0, i_0, e_{p,0}, \lambda_0)^\top. \quad (8)$$

The connection passes one block's output into the other block's equation. Their derivatives and the four initial coordinates remain part of the assembled system.

3 The same connection using phasors

Use the template's convention $x(t) = \text{Re}(\hat{x}e^{i\omega t})$, $\omega > 0$, so $D \mapsto i\omega$. The two local equations give

$$\begin{aligned} Z_s(\omega) &= i\omega L_s + R_s + \frac{1}{i\omega C_s}, & \hat{e}_s &= Z_s \hat{i}, \\ Y_p(\omega) &= i\omega C_p + \frac{1}{R_p} + \frac{1}{i\omega L_p}, & \hat{i} &= Y_p \hat{e}_p. \end{aligned} \quad (9)$$

Here Z_s multiplies current to give voltage, while Y_p multiplies voltage to give current. The connection equations retain their form:

$$\hat{e} = Z_s \hat{i} + \hat{e}_p, \quad \hat{i} = Y_p \hat{e}_p. \quad (10)$$

Substituting the second into the first gives

$$\Delta(\omega) = 1 + Z_s Y_p, \quad \boxed{\hat{e}_p = \frac{\hat{e}}{\Delta}, \quad \hat{i} = \frac{Y_p \hat{e}}{\Delta}, \quad \Delta \neq 0.} \quad (11)$$

The internal coordinate phasors are $\hat{q} = \hat{i}/(i\omega)$ and $\hat{\lambda} = \hat{e}_p/(i\omega)$. Within the parallel block the three currents are

$$\hat{i}_C = i\omega C_p \hat{e}_p, \quad \hat{i}_R = \frac{\hat{e}_p}{R_p}, \quad \hat{i}_L = \frac{\hat{e}_p}{i\omega L_p}, \quad \hat{i}_C + \hat{i}_R + \hat{i}_L = \hat{i}. \quad (12)$$

These formulas describe the harmonic component. Initial coordinates for the full time-dependent solution remain those of (7).

3.1 Combining further blocks

For any series group with $\hat{e}_j = Z_j \hat{i}$, summation gives $Z_{eq} = \sum_j Z_j$. For any parallel group with $\hat{i}_j = Y_j \hat{e}$, summation gives $Y_{eq} = \sum_j Y_j$. The individual phasors follow as

$$\hat{e}_j = \frac{Z_j}{\sum_\ell Z_\ell} \hat{e} \quad (\text{series}), \quad \hat{i}_j = \frac{Y_j}{\sum_\ell Y_\ell} \hat{i} \quad (\text{parallel}). \quad (13)$$

These quotients require nonzero denominators. Where finite and nonzero, $Y = 1/Z$ converts between the two descriptions, so a nested series/parallel group can be combined from its inner branches outward. Sum complex phasors before taking magnitudes. At a zero denominator, retain the original linear equations and check their compatibility.

4 Eliminating the internal coordinates

For a scalar reduction, assume that $e(t)$ is twice continuously differentiable. Define the two local polynomials

$$A(s) = L_s s^2 + R_s s + \frac{1}{C_s}, \quad B(s) = C_p s^2 + \frac{s}{R_p} + \frac{1}{L_p}. \quad (14)$$

Equations (5)–(6) read $A(D)q + D\lambda = e$ and $B(D)\lambda = Dq$. Apply $B(D)$ to the first. Constant coefficients allow $B(D)$ and D to commute, giving

$$\boxed{K(D)q = B(D)e, \quad K(s) = A(s)B(s) + s^2.} \quad (15)$$

The fourth-order polynomial is

$$\begin{aligned} K(s) &= L_s C_p s^4 + \left(\frac{L_s}{R_p} + R_s C_p \right) s^3 \\ &\quad + \left(\frac{L_s}{L_p} + \frac{R_s}{R_p} + \frac{C_p}{C_s} + 1 \right) s^2 \\ &\quad + \left(\frac{R_s}{L_p} + \frac{1}{C_s R_p} \right) s + \frac{1}{C_s L_p}. \end{aligned} \quad (16)$$

Its product term comes from the two local ODEs; the additional s^2 comes from connecting their terminal derivatives.

Working with ODE blocks also makes higher-order extensions easier to assemble where suitable. Construct coefficients using *ODE Coefficient Synthesis*, Sections 1–4, and insert them into a block's local equation. The series/parallel connection rules remain unchanged, avoiding repeated branch-by-branch current and voltage derivations. For phasors, substitute $D = i\omega$ in the extended polynomial. Any increase in differential order requires the corresponding initial data, consistent with the connection constraints.

For the mixed connection above, the eliminated variables are reconstructed by

$$i = Dq, \quad e_p = e - A(D)q, \quad \lambda = L_p \left(i - C_p D e_p - \frac{e_p}{R_p} \right). \quad (17)$$

Equation (15) implies $B(D)e_p = D^2q$; differentiating the last formula then gives $D\lambda = e_p$. Thus reconstruction satisfies the original coupled equations.

To match the initial state in (8), set $q(t_0) = q_0$ and $(Dq)(t_0) = i_0$, then use

$$\begin{aligned} (D^2q)(t_0) &= \frac{e(t_0) - R_s i_0 - q_0/C_s - e_{p,0}}{L_s}, \\ (D^3q)(t_0) &= \frac{(De)(t_0) - R_s (D^2q)(t_0) - i_0/C_s - (De_p)(t_0)}{L_s}, \end{aligned} \quad (18)$$

where $(De_p)(t_0) = (i_0 - e_{p,0}/R_p - \lambda_0/L_p)/C_p$. The four scalar initial values therefore encode the same four coordinates as the coupled system.

Finally, $Z_s = A(i\omega)/(i\omega)$ and $Y_p = B(i\omega)/(i\omega)$ imply

$$K(i\omega) = (i\omega)^2 \Delta(\omega). \quad (19)$$

Consequently $\hat{q} = B(i\omega)\hat{e}/K(i\omega)$, together with (17), gives exactly (11) whenever its denominator is nonzero. The scalar ODE and the phasor formulas are two reductions of the same local equations and connection constraints. The additional step from signed voltage-current products to work is developed in *Energy Ledgers for Forced Harmonic ODEs*, Parts 1 and 2.

References

1. H. Nilre and B. C. Herlin (2026). *The ODE Template and Its Domain Equivalents*.
2. H. Nilre and B. C. Herlin (2026). *ODE Coefficient Synthesis*.
3. H. Nilre and B. C. Herlin (2026). *Energy Ledgers for Forced Harmonic ODEs*.